5(a)	Any two of the following: ● Energy is transferred along the direction of propagation of progressive waves while there is no propagation of energy in a stationary wave. ● There is a propagation of the waveform (the crests/troughs/compressions/rarefactions of the waves) in a progressive wave but the waveform of a stationary wave does not move but instead undergoes changes in shape. ● All particles within an internodal segment are in phase in a stationary wave while the phase difference of particles in progressive waves is proportional to their distance apart. ● Amplitudes of vibrations of particles in a stationary wave vary from a minimum at the nodes to a maximum at the antinodes while the amplitude of a progressive wave (propagating in a plane) remains the same.	B2
(b)(i)	A (micro)wave (from the transmitter) is <u>incident on the sheet</u> and <u>reflects</u> at Y The <u>incident and reflected waves overlap and superpose</u> to form stationary wave.	B1 B1
(b)(ii)	Distance between PY = 50 x (0.5 λ) 1.5 = 50 x (0.5 λ) λ = 0.060 m	M1 A1
(b)(iii)(1)	<u>wavelength decreases,</u> distance between maxima/minima decreases Number of maximum amplitude increases	M1 A1
(b)(iii)(2)	<u>wavelength is unchanged</u> so (PQ is) the same (as QR). Distance between maxima/minima remains the same. Number maximum amplitude remains unchanged	M1 A1
